

Design of the Seated Navigation for Immersive Lower Limb Exergame

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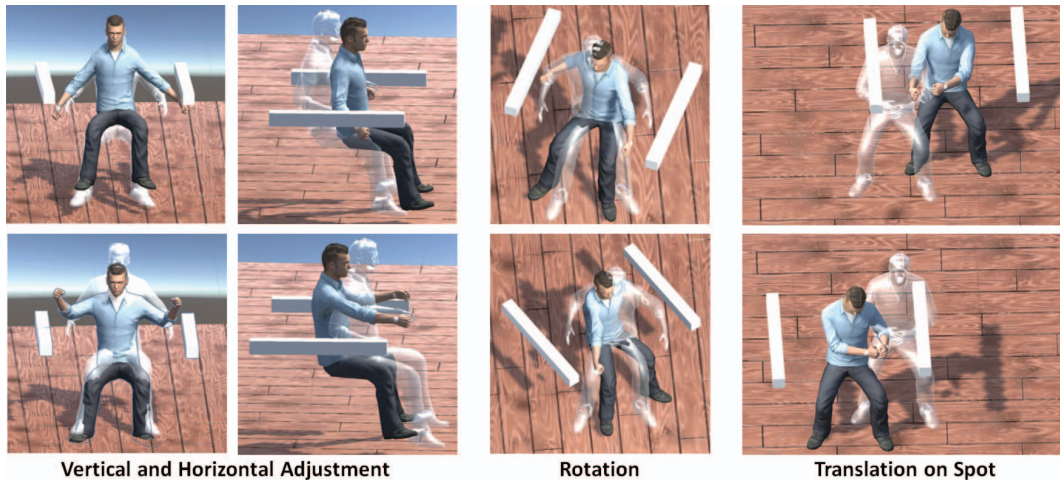


Figure 1: Dual Point-Tugging: the transparent shape is the original sitting pose, and the textured avatar [4] is the updated pose.

ABSTRACT

Immersive lower limb exergames have been used to motivate physical training or rehabilitation. Due to the relative constraint on mobility in the seated pose, interacting with the distanced object using the leg could be challenging. To tackle the challenge, we describe the detailed design and implementation of Dual Point-Tugging (DPT) seated navigation. The DPT interface unifies the translation and rotation, allowing users to refine their position and extend leg reachability in the immersive virtual environment. Also, by rearranging the arm position, the user could continuously perform swift translation around a spot in a comfortable and balanced pose.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Engaging gaming experiences within Immersive Virtual Environments (IVEs) has been used to motivate physical training or rehabilitation [1]. The immersive exergame could target lower limb activity by integrating a motion capture system with the Head-Mounted Display (HMD). In order to avoid falling or imbalance during potentially intense lower exertion, a seated pose could be a safer option. However, gaming engagement might be lower than standing since the user may feel more passive in a seated pose. Without the ability to walk around physically, the safe reachability of the lower limb in IVE could be more restricted in a seated pose [6].

The ability to navigate through the IVE in a seated pose can safely increase the reachability of the user's lower limbs. Plus, to reach the target on the virtual ground using leg movement without being blocked by the physical floor, the ability to precisely adjust the vertical position is essential [2]. Therefore, the navigating interface

for the lower limb exergame should handle movements at least in four degrees of freedom: forward/backward, up/down, sideways, and yaw rotation.

Using the hand-held controller, the user can highly engage in the seated arm-swinging approach since a significant amount of body movement is involved [3]. However, arm-swinging only allows moving in a single direction if the user is seated on a fixed chair. Point-tugging enables users to move towards different directions yet does not allow users to rotate [3]. Leaning-based approaches such as NaviChair [5] further provide vestibular cues to reduce motion sickness. However, leaning-based methods may interfere with the lower body movements needed for interaction with game objects. On the other hand, joystick-based navigation could be the most effortless approach to navigating precisely in the IVE. Nevertheless, without sufficient motion cues, the user may experience less embodiment and suffer a higher degree of motion sickness [6].

2 PROPOSED NAVIGATION FOR LOWER LIMBS EXERGAME

2.1 Design Logic

As a reachability enhancement for lower limb exergames, we tried to minimize the interference on the leg movement during navigation so that the navigation can happen with any leg movement at the same time if needed. Hence, controllers on the hands will be an excellent interface to handle the locomotion in IVE, as the legs are busy playing the game.

Inspired by the Point-Tugging navigation, this work introduces Dual Point-Tugging (DPT) to incorporate the rotation into a unified interface further. DPT is a continuous motion-based navigation that leverages a pair of lateral virtual handles around the avatar for both controllers to grab (as shown in Fig. 1). In addition to precise adjustment, swiftly moving around to dodge or hit a target timely could be a desired feature for a higher pace exergame. Therefore, a translation on-spot operation is integrated into the design of DPT.

2.2 Usage and Implementation

To activate the DPT navigation, users first need to have both hands hold the virtual handles on lateral sides, just like people adjust their sitting pose using a chair armrest. Once activated, the DPT will stay

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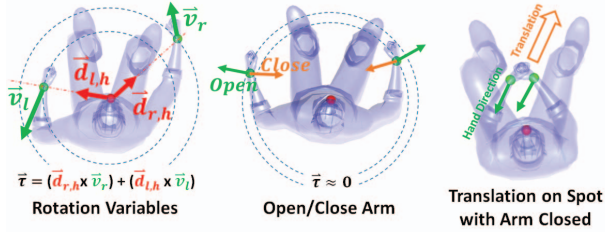


Figure 2: Rotation variables with its calculation on $\vec{\tau}$ and open/close pose

working until the holding button on one of the controllers is released. The detail of DPT is described below, and the implementation can be found in Algorithm 1.

2.2.1 Steady Translation

Both hands need to pull or push toward a similar direction to move their avatar towards the opposite direction. The similarity could be evaluated based on the dot product of both directions. Vertical adjustment is separated from the horizontal one since the vertical one will become less important once the user has been adjusted for reaching the target on the virtual floor properly. Accordingly, DPT requires the user's both hands to move close enough to the vertical direction for moving vertically. Otherwise, the user will move horizontally.

2.2.2 Rotation

For rotation, the design of DPT is inspired by a manual wheelchair. To rotate a manual wheelchair, both hands need to rotate the wheels in the opposite direction. Then the wheelchair will rotate based on the torque contributed from both sides. Similarly, the virtual avatar will rotate on the yaw axis in DPT if both hands hold the virtual handle move opposite and contribute to the same rotation direction. For example, the avatar will yaw counterclockwise if the user's right hand moves forward and the left hand moves backward.

2.2.3 Swift Translation on Spot

For swift translation around on the spot, the user could close both arms to the front of the chest. Such arms' closing movement does not cause locomotion since the hands do not move in the same direction for translation nor generate consistent torque for rotation. With the arms rearrangement to the front of the chest, the user could be more relaxed for continuous usage since they no longer keep holding their arms wide open in the air.

Moreover, the translation on the spot may help keep the user in a better-balanced pose while exergaming with the leg. Users might over-stretch their legs to reach a distanced target in IVE. However, this action may shift their body mass center forward and increase the potential risk of falling. With translation on the spot, the user could reach farther by moving their hand in the opposite direction. This might help to reduce the potential body mass center shifting issue. Lastly, the user will move back to the initial spot when they pull the arms back to the front of the chest.

2.3 Potential Improvements

The virtual handles on the lateral sides provide visual feedback of the DPT. While using the DPT in an exergame, those handles might not be noticeable if users focus on targets in front of themselves. Hence, a state change on the virtual hand (e.g., showing an object is held on hand) could be used to remind the user whether the DPT is activated.

The handles' distance from the body could be a parameter to tune. If the virtual handles are very close to the body, the user could easily access it repeatedly for long-distance cruising. However, the handles could also interfere virtual hand's interaction with other objects nearby. On the contrary, if the handles are far enough, it

Algorithm 1 Implementation of DPT

Require:

- Vectors below are in avatar's coordinate (left:x, up:y, front:z)
- $\vec{v}_l, \vec{v}_r$ $\triangleright$ Left/Right controller's velocity at the frame
- $\vec{d}_{l,h}, \vec{d}_{r,h}$ $\triangleright$ Unit vectors representing the horizontal direction from the HMD to the left/right controller (Fig. 2)
- $\vec{v}_{up}$ $\triangleright$ Unit vector toward the up direction
- Δt $\triangleright$ Time difference since the last render update
- $s_v = .1, s_h = .5$ $\triangleright$ Speed of vertical/horizontal movement
- $t_v = .7, t_h = .01$ $\triangleright$ Vertical/horizontal movement threshold
- $s_{rot} = 50, t_{rot} = .01$ $\triangleright$ Speed/threshold of rotation

procedure DPTMOVE

$\triangleright$ Perform for every frame

if not both controller in engaged state **then**

return

if $(|\vec{v}_l \cdot \vec{v}_r| \cdot \text{norm.y} + |\vec{v}_r \cdot \vec{v}_r| \cdot \text{norm.y}) / 2 > t_v$ **then** $\triangleright$ vertical

$\vec{v}_o = s_v(\vec{v}_l + \vec{v}_r)$; $\vec{v}_o.x = 0$; $\vec{v}_o.z = 0$

translate to $-\Delta t \vec{v}_o$

else if $|\vec{v}_l \cdot \vec{v}_r| > t_h$ **then** $\triangleright$ horizontal

$\vec{v}_o = s_h(\vec{v}_l + \vec{v}_r)$; $\vec{v}_o.y = 0$

translate to $-\Delta t \vec{v}_o$

else if $|\vec{v}_l \cdot \vec{v}_r| < -t_{rot}$ **then** $\triangleright$ rotation

$\vec{\tau}_l = \vec{d}_{l,h} \times \vec{v}_l$; $\vec{\tau}_r = \vec{d}_{r,h} \times \vec{v}_r$

$a_{rot} = s_{rot}((\vec{\tau}_l + \vec{\tau}_r) \cdot \vec{v}_{up})$

$\triangleright a_{rot}$ will close to 0 when open/close arm (Fig. 2)

yaw rotate $\Delta t a_{rot}$ degrees

would interfere less with accessing nearby objects. However, it would require users to put more effort into reaching them and could be a hassle for cruising.

3 CONCLUSION AND FUTURE WORKS

This work discloses the design and implementation of the DPT technique. The DPT interface unifies the translation and rotation by extending the point-tugging. Further, an arm adjustment strategy for moving in a comfortable pose is included in the design of the interface. In the future, a series of user experiments will be needed to examine the proposed design thoroughly.

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